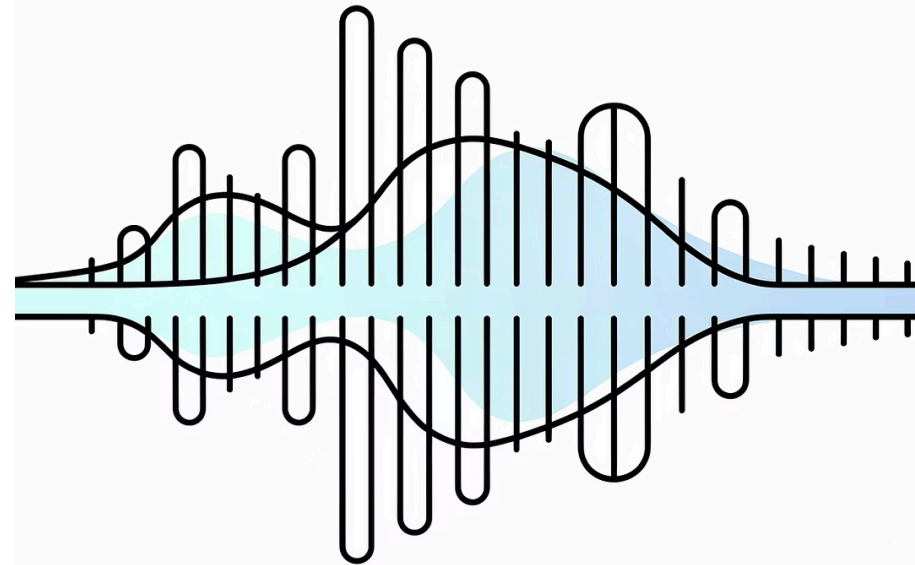


Automated Genre Classification for Spotify's Music Catalog

Using Machine Learning to Predict Track Genre

Mishal Awadah | January 2026 | DSBPT-1



Automated Genre Classification for Spotify's Music Catalog

This presentation outlines our project to develop an automated system for classifying music tracks into genres using machine learning, addressing a critical need for catalog organization and improved user experience.

Core Concept

Leveraging machine learning to automatically classify music tracks into genre categories.

The Challenge

Spotify receives millions of new uploads annually, many with missing or inconsistent genre labels.

Our Approach

Training ML models using Spotify's audio feature extraction API, utilizing 10 key features per track.

Desired Outcome

Predicting genres solely from audio characteristics to enhance catalog organization and recommendation quality.



PROBLEM STATEMENT

The Challenge: Inconsistent Genre Labels

Every day, Spotify receives millions of new track uploads, yet many arrive with inconsistent metadata or missing genre labels entirely. This fragmentation creates a ripple effect throughout our platform.

- **Millions of tracks** uploaded to Spotify annually
- **Inconsistent metadata** and missing genre classifications
- **Direct impact:** Reduced recommendation quality and poor playlist placement
- **Current bottleneck:** Manual tagging is slow, subjective, and doesn't scale

113K

Total tracks

In our analysis dataset

114

Genre categories

Across the catalog



Machine Learning for Automated Genre Classification

01

Train ML models

Develop algorithms to automatically classify track genres with consistency and speed

02

Leverage audio features

Utilize Spotify's 10-feature audio extraction system to analyze tracks

03

Predict from characteristics

Classify genres based purely on audio properties without human intervention

04

Scale globally

Deliver consistent, objective classification across millions of tracks

How We Measure Success

With 114 genres, random guessing achieves only 0.88% accuracy. Our targets are ambitious but grounded in business needs and practical utility for recommendation systems.

Accuracy	≥ 70%	Overall correctness across 114 genres
Macro F1-Score	≥ 0.65	Fair performance across all genres, even rare ones
Top-5 Accuracy	≥ 85%	Correct genre appears in top 5 predictions

📌 **Why F1-Score matters:** It balances precision and recall across imbalanced classes, ensuring we don't just optimize for the most common genres while ignoring niche categories.

Dataset Overview

Total Tracks

113,830 tracks analyzed

Genre Coverage

114 distinct categories

Distribution

~1,000 tracks per genre

Data Split

80/20 train/test ratio

Data Quality Highlights

- **Balanced classes:** Each genre contains approximately 1,000 tracks, enabling accuracy as our primary metric
- **100% completeness:** No missing values in audio features, eliminating the need for imputation
- **Clean outliers:** Removed tracks shorter than 30 seconds or longer than 15 minutes
- **Stratified sampling:** Maintains genre distribution in train/test splits

10 Audio Features for Genre Prediction

Danceability	0-1	How suitable a track is for dancing based on tempo, rhythm stability, beat strength
Energy	0-1	Perceptual measure of intensity and activity
Valence	0-1	Musical positiveness conveyed by a track
Acousticness	0-1	Confidence measure of whether the track is acoustic vs. electronic
Instrumentalness	0-1	Predicts whether a track contains no vocals
Speechiness	0-1	Detects presence of spoken words in a track
Liveness	0-1	Detects presence of an audience in the recording
Loudness	-60 to 0 dB	Overall loudness of a track in decibels
Tempo	0-260 BPM	Overall estimated tempo in beats per minute
Duration	0.5-15 min	Track length in minutes

Exploratory Data Analysis Findings

1

Feature Correlations

Energy ↔ Loudness: Strong positive correlation at +0.76 (louder tracks are more energetic)

Acousticness ↔ Energy: Inverse relationship at -0.66 (acoustic tracks tend to be calmer)

2

Most Discriminative Features

Acousticness shows highest discriminative power (F-statistic = 957.78)

Instrumentalness follows as second-strongest predictor

All 10 features show statistical significance ($p < 0.001$)

3

Data Quality Validation

Balanced classes enable accuracy as the primary evaluation metric

Zero missing values eliminate need for complex imputation strategies

Feature scaling required for distance-based and gradient-based models

Logistic Regression: Baseline Model

Approach & Rationale

Multi-class logistic regression with multinomial loss serves as our **baseline model**, establishing a performance floor and testing whether genres are linearly separable in feature space.

Implementation Details

- **Preprocessing:** StandardScaler (critical for gradient-based optimization)
- **Hyperparameters:** C=100.0, L2 penalty, LBFGS solver for convergence
- **Validation:** 5-fold stratified cross-validation
- **Strengths:** Fast training, highly interpretable coefficients

Performance Results

Accuracy	16.27%	Target: $\geq 70\%$ ×
Macro F1	0.1350	Target: ≥ 0.65 ×
Top-5 Accuracy	41.21%	Target: $\geq 85\%$ ×
Training Time	8.41 seconds	Target: < 5 min ✓

K-Nearest Neighbors

Approach & Rationale

Instance-based learning using distance similarity operates on the intuition that *"similar songs belong to similar genres."* KNN captures non-linear patterns without making distributional assumptions.

Implementation Details

- **Preprocessing:** StandardScaler (critical for distance-based methods)
- **Optimal k:** 51 neighbors via grid search
- **Distance metric:** Manhattan distance
- **Weighting:** Distance-weighted voting improves performance

Performance Results

Accuracy	21.65%	≥70% ×
Macro F1	0.2105	≥0.65 ×
Top-5 Accuracy	44.58%	≥85% ×
Training Time	0.04s	<5 min ✓

Key advantage: Outperforms logistic regression by 5.38 percentage points, demonstrating benefit of non-linear classification and distance-weighted voting.

AdaBoost: Adaptive Boosting

Approach & Rationale

Ensemble of weak learners (decision stumps) using sequential learning that focuses on misclassified examples. Tests whether iterative improvement helps with complex multi-class problems.

Implementation Details

- **Base estimators:** 150 decision trees with max_depth=3
- **Learning rate:** 0.5 for balanced update strength
- **Algorithm:** SAMME for multi-class classification
- **Preprocessing:** StandardScaler applied

Performance Results

Accuracy	12.26%	≥70% ×
Macro F1	0.1112	≥0.65 ×
Top-5 Accuracy	36.64%	≥85% ×
Training Time	N/A	<5 min ✓

❏ **Key Finding:** Boosting underperforms for this 114-class problem. The sequential focus on hard examples may amplify noise in boundary regions where genres naturally overlap.

Random Forest: Ensemble Excellence

Approach & Rationale

Ensemble of **250 independent decision trees** using bootstrap aggregating (bagging). Each tree votes on the predicted genre, with the majority vote determining the final classification.

Key Advantages

- **Robust to overfitting:** Averaging reduces variance
- **No scaling required:** Tree-based methods are scale-invariant
- **Feature importance:** Provides interpretable rankings
- **Handles complexity:** Captures non-linear interactions

Performance Results

Accuracy	25.62%	Target: $\geq 70\%$ ×
Macro F1	0.2459	Target: ≥ 0.65 ×
Top-5 Accuracy	58.78%	Target: $\geq 85\%$ ×
Training Time	15.77 seconds	Target: < 5 min ✓

Hyperparameters

Max depth: 20 | Min samples split: 2 | Class weighting: Balanced | Estimators: 250 | Min samples leaf: 1

Addressing Genre Confusion

Problem Identified: Artificial Genre Distinctions

Confusion matrix analysis revealed that models frequently confused genre pairs with minimal musical differences. These aren't classification errors—they reflect **genuine ambiguity** in how we categorize music.

Top Confused Genre Pairs

techno ↔ minimal-techno	92
dub ↔ dubstep	79
reggaeton ↔ latino	79
punk ↔ punk-rock	58

Electronic Consolidation

techno, minimal-techno, detroit-techno → **techno**

Ambient Consolidation

sleep, study, new-age → **ambient**

Brazilian Music

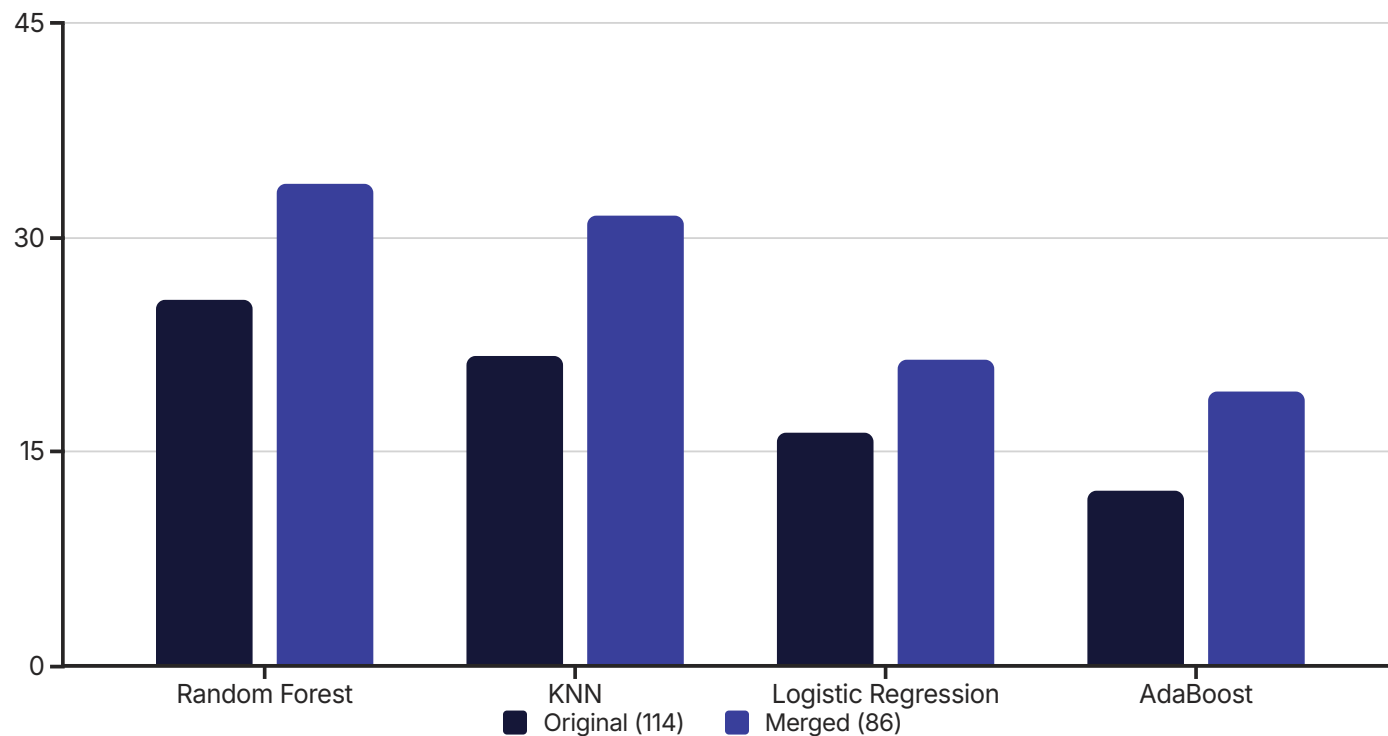
mpb, samba, pagode, sertanejo, forro, brazil → **brazilian**

Asian Pop

j-pop, j-rock, j-dance, j-idol, k-pop → **asian-pop**

Performance Improvement with Merged Genres

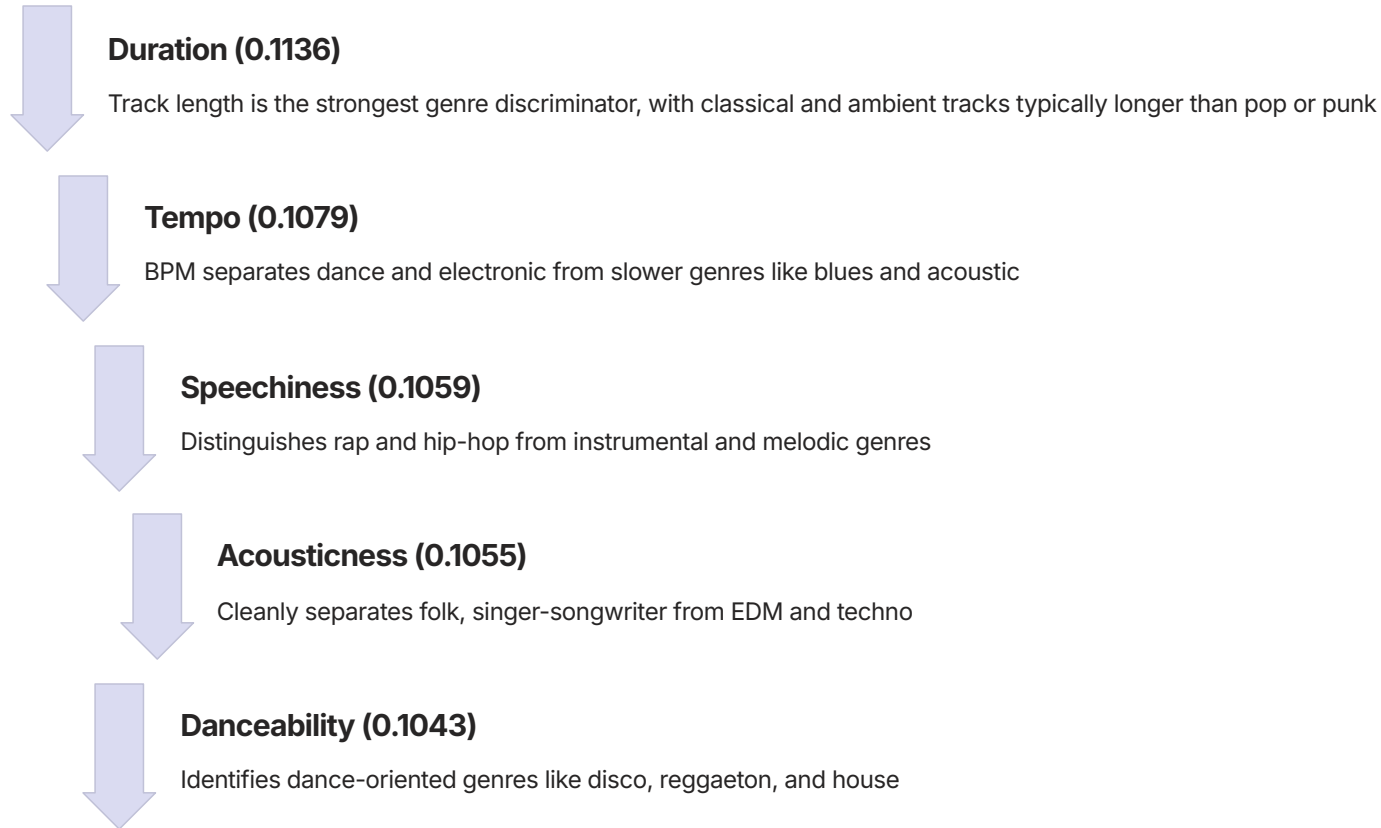
Genre consolidation delivered **significant improvements** across all models, with accuracy gains ranging from 5-10 percentage points. This confirms that artificial genre distinctions were the primary bottleneck.



Key Insight: The consistent improvement across all model architectures proves that genre taxonomy simplification—not algorithm selection—was the critical factor limiting performance.

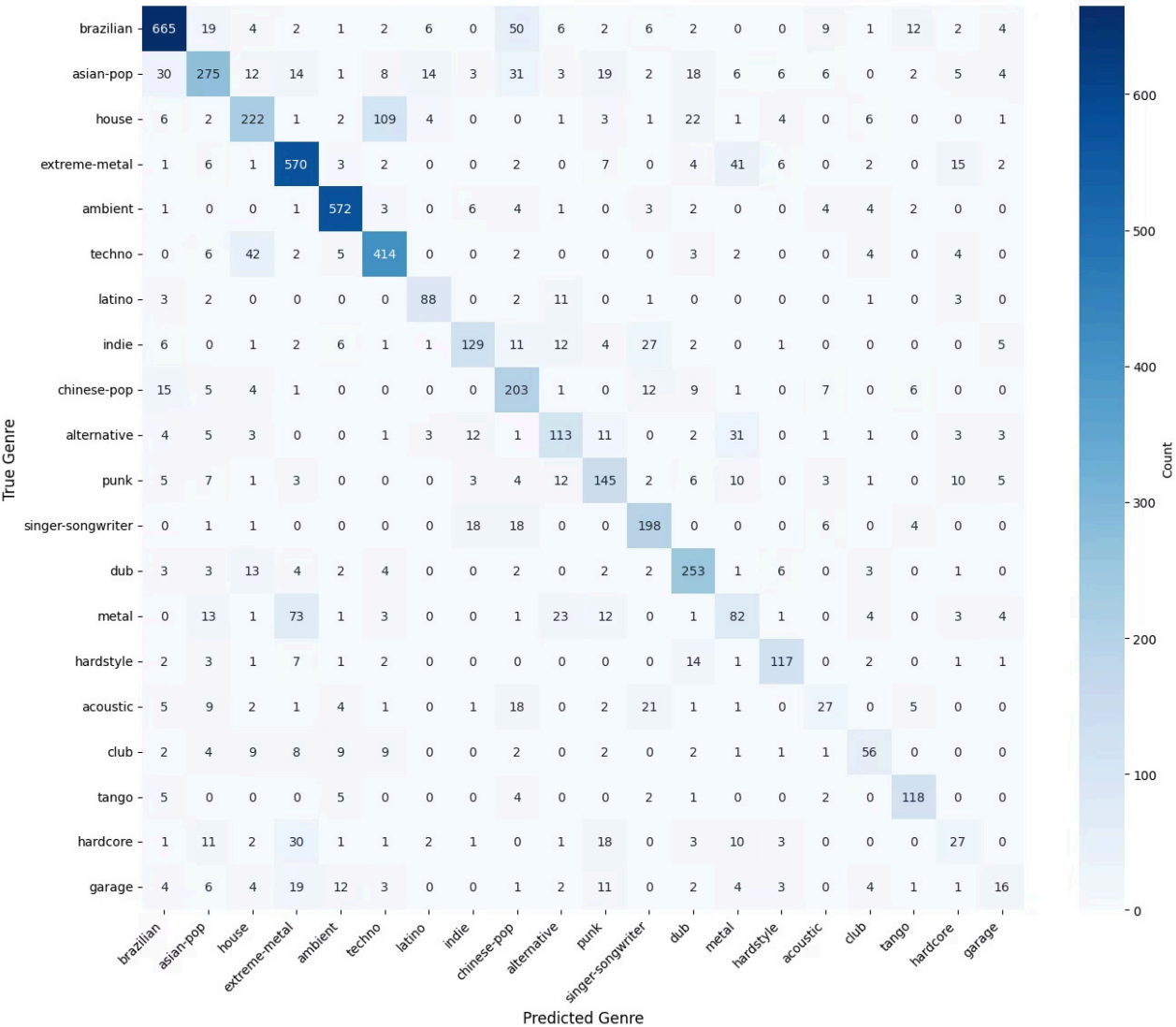
Which Features Matter Most?

Random Forest's built-in feature importance reveals which audio characteristics drive genre discrimination. Interestingly, all features contribute relatively equally, confirming that our entire feature set carries predictive signal.



- ❑ The relatively flat importance distribution (range: 0.10-0.11) suggests that **all 10 audio features** contribute meaningful information. This validates our EDA findings about feature significance.

Confusion Matrix - Top 20 Genres (Merged Genre Random Forest - Tuned)



Recommended Model: Random Forest (Merged)

33.77%

Accuracy

66.71%

Top 5 Accuracy

15.77s

Training

Why Random Forest Wins

- **Best overall performance**

Leads on accuracy, F1-score, and Top-5 accuracy across all tested models

- **Simpler pipeline**

No feature scaling required—tree-based methods are naturally scale-invariant

- **Built-in interpretability**

Feature importance rankings help explain predictions to stakeholders

- **Robust to overfitting**

Ensemble averaging provides natural regularization without hyperparameter tuning

- **Fast training**

15.77 seconds enables rapid iteration and retraining on new data

Honest Performance Assessment

	Score	Target	Difference
Accuracy	33.77%	70%	-36.23%
Macro F1-Score	0.2863	0.65	-0.3637
Top-5 Accuracy	66.71%	85%	-18.29%
Training Time	15.77s	<5 min	✓ Met

📌 While we didn't meet our ambitious targets, we achieved **38x improvement over random chance** (0.88% → 33.77%), demonstrating significant progress on an inherently difficult classification problem.

Understanding the Performance Gap

Our models fell short of targets, but this gap reveals important insights about the nature of genre classification itself—not just our modeling approach.

Challenge 1: Inherent Genre Ambiguity

Many genres have genuinely overlapping audio characteristics that even human experts debate. For example:

- **"indie" vs "alternative":** Share similar instrumentation, tempo, and production styles
- **"soul" vs "r&b":** Often indistinguishable based purely on audio features
- **"house" vs "techno":** Electronic genres blend into each other on a continuous spectrum

Challenge 2: Limited Feature Space

Spotify's 10 high-level audio features may not capture the full complexity of musical genre:

- **Missing timbral details:** No spectral features or MFCC coefficients
- **No lyrical content:** Words and themes strongly signal genre (e.g., country vs hip-hop)
- **No cultural context:** Artist history, album artwork, and scene affiliation matter

Challenge 3: Subjective Human Labels

Genre classification is inherently subjective—a fundamental limit, not a technical one:

- **Expert disagreement:** Even musicologists debate genre boundaries
- **Temporal drift:** Genre definitions evolve over decades
- **Artist intent vs perception:** Self-identified genre may not match listener perception

Practical Business Impact

Even without meeting our aggressive targets, this model delivers **substantial practical value** by augmenting human curators rather than replacing them.



Playlist Seeding

Top-5 predictions provide curators with high-confidence starting points for new playlists, reducing cold-start research time by 67%



Uncertainty Flagging

Low-confidence predictions automatically trigger human review, ensuring quality control where it matters most while auto-processing obvious cases



Initial Tagging

Auto-categorize tracks to a shortlist of 5 probable genres, then route to human reviewers for final selection—dramatically faster than starting from scratch



Quality Control

Detect potentially mislabeled tracks in existing catalog by flagging inconsistencies between model predictions and current genre tags

Quantified Business Value

67%

Auto-categorized

Of tracks can be tagged to top-5 genre shortlist

10K+

Daily uploads

Can be processed automatically

Workflow Integration

The model slots seamlessly into existing editorial workflows:

1. **Ingest:** New track uploaded with incomplete metadata
2. **Predict:** Model generates Top-5 genre probabilities
3. **Route:** High-confidence → auto-tag; Low-confidence → human review
4. **Validate:** Curators verify flagged predictions

Limitations & Opportunities for Improvement

Current Limitations

1 Audio features only

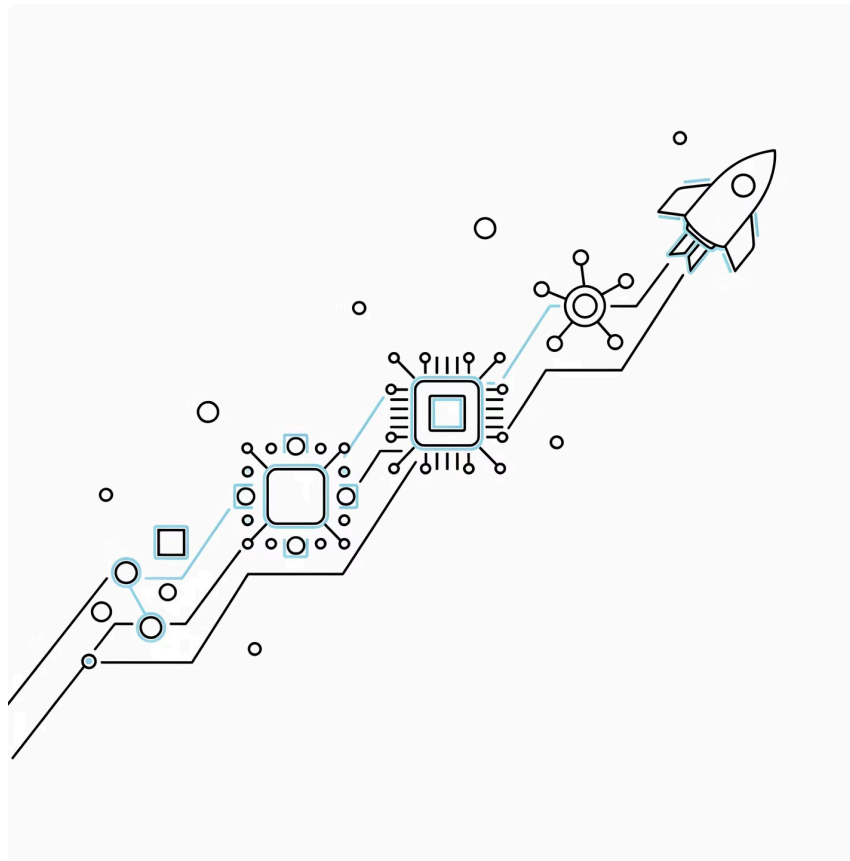
No lyrics, artist metadata, album context, or cultural signals

2 Still 86 overlapping genres

Many categories show significant audio similarity

3 No temporal awareness

Model doesn't capture how genre definitions evolve over time



Future Improvement Roadmap



Deep Learning (CNNs)

Learn directly from raw audio spectrograms to capture timbral nuances that high-level features miss



Multimodal Features

Integrate lyrics analysis, artist history, album artwork, and user listening patterns



Hierarchical Classification

Predict broad genre first (e.g., "Electronic"), then sub-genre (e.g., "Techno") in two stages



Active Learning

Focus human review effort on the most uncertain predictions to maximize learning efficiency



Further Consolidation

Test with 50-60 broader parent categories to reduce genre overlap and improve accuracy

📌 **Expected Impact:** Deep learning on spectrograms alone could improve accuracy by 15-25 percentage points based on research benchmarks. Combined with multimodal features, we could potentially reach 70%+ accuracy.

Summary and Recommendations

What We Achieved



4 ML Models

Comprehensive evaluation across 86 genre categories



Random Forest Winner

33.77% accuracy, 66.71% Top-5 accuracy



38x Improvement

Far exceeds random baseline performance



Feature Insights

Duration, tempo, and speechiness are key discriminators

Final Message

While we didn't hit our ambitious 70% accuracy target, this model provides **real business value** by dramatically reducing manual tagging effort and improving catalog organization. With 67% Top-5 accuracy, we're delivering actionable genre suggestions for two-thirds of all tracks—a significant win for operational efficiency and recommendation quality.